

# AI Financial Advisor Platform

## Professional MERN Stack Project Scope

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This document presents the professional-level project scope for an AI-powered Financial Advisor Platform built using the MERN stack. The system combines financial management tools, intelligent analytics, AI-powered recommendations, and SaaS architecture concepts to create a modern FinTech application suitable for portfolio, academic, and industry-level development.

### 1. Project Overview

The platform helps users manage expenses, track budgets, monitor savings goals, receive AI-based financial insights, and generate financial reports. It acts as a personal finance assistant capable of analyzing user spending behavior and providing smart recommendations.

### 2. Objectives

- Build a scalable FinTech platform using MERN stack.
- Integrate AI for personalized financial insights.
- Provide analytics dashboards and forecasting systems.
- Implement secure authentication and role-based access.
- Demonstrate enterprise-grade SaaS architecture.

### 3. User Roles

**User:** Manage finances, track expenses, receive AI insights.

**Admin:** Manage users, monitor subscriptions, analyze platform usage.

### 4. Core Features

#### Authentication System

- JWT authentication
- Refresh token implementation
- Google OAuth login
- Email verification

- Password reset system

### **Expense & Income Tracking**

- Add/edit/delete transactions
- Categorize expenses
- Recurring transactions
- Search & filters

### **Budget Management**

- Monthly budget setup
- Category-wise budget limits
- Overspending alerts

### **AI Financial Advisor**

- AI-powered chatbot
- Expense analysis
- Savings recommendations
- Financial forecasting

### **Analytics Dashboard**

- Expense graphs
- Income vs expense comparison
- Savings analytics
- Financial trends visualization

## **5. Advanced AI Features**

- AI expense prediction
- Personalized budgeting recommendations
- Investment guidance based on risk profile
- Fraud/anomaly detection
- AI-generated financial summaries
- OCR-based receipt scanning

## 6. Technology Stack

**Frontend:** React.js / Next.js, Tailwind CSS, Redux Toolkit, React Query

**Backend:** Node.js, Express.js

**Database:** MongoDB, Mongoose

**AI:** OpenAI API / Gemini API, LangChain

**Realtime:** Socket.IO

**Deployment:** Vercel, Render, AWS

## 7. Database Collections

- Users
- Transactions
- Budgets
- Goals
- Investments
- Notifications
- AIChats
- Reports
- Subscriptions

## 8. Security Features

- Password hashing using bcrypt
- Protected routes with JWT
- Rate limiting
- Input validation
- Secure API architecture
- Environment variable management

## 9. SaaS Features

- Subscription plans
- Razorpay/Stripe integration
- Premium AI tools
- Usage monitoring
- Revenue analytics dashboard

## 10. Future Enhancements

- AI voice assistant
- Mobile application
- Bank API integration
- Smart tax planning
- Multi-currency support
- Advanced portfolio management

## 11. Project Outcomes

This project demonstrates expertise in full-stack development, AI integration, real-time communication, dashboard analytics, SaaS architecture, authentication systems, and enterprise-grade backend design. It is highly suitable for professional portfolios, job interviews, internships, and startup showcases.

## 12. Recommended Architecture

| Layer       | Technology            |
|-------------|-----------------------|
| Frontend    | React.js / Next.js    |
| Backend     | Node.js + Express.js  |
| Database    | MongoDB + Mongoose    |
| AI Services | OpenAI / Gemini APIs  |
| Realtime    | Socket.IO             |
| Deployment  | Vercel / AWS / Render |